

Maintenance Wizard Report

Diagnosis

Severe degradation and imminent failure of the CCM Segment Bearing, likely due to internal damage.

Root Cause

Probable root causes include lubrication failure (e.g., grease washout, blocked lines, inadequate re-lubrication), contamination ingress (water, scale), excessive load, or fatigue. The combination of symptoms suggests advanced internal wear.

Risk Level

High

Failure Probability

Very High / Imminent (within hours to days under continuous operation)

Remaining Useful Life

Critically low; estimated less than 24-48 hours under current operating conditions before catastrophic failure.

Catastrophic Risk

High. Potential for strand breakage, severe damage to the caster segment, rolls, and associated machinery. Significant production loss due to extended unplanned downtime and safety hazards from component failure.

Immediate Actions

['Notify Production/Operations management immediately about the critical condition.', 'Continuously monitor vibration and temperature trends on the affected segment/bearing.', 'Investigate lubrication lines to the bearing for blockage or damage; attempt to re-lubricate if possible and safe.', 'Prepare all necessary tools, personnel, and spare parts for an immediate segment change-out or bearing replacement.', 'Evaluate possibility of reducing casting speed or load to mitigate failure progression if operation must continue temporarily.', 'Schedule an immediate planned shutdown for segment replacement at the earliest safe opportunity.']

Short Term Actions

['Execute segment replacement (or bearing replacement if segment change is not feasible) during the next available maintenance window or forced outage.', 'Collect detailed operational data (casting speed, temperature, vibration) leading up to the failure.', 'Isolate and tag the failed bearing/segment for detailed post-mortem analysis.', 'Inspect adjacent bearings and lubrication']

systems on the same segment and neighboring segments for similar signs of wear or issues.】

Long Term Actions

【Conduct a thorough root cause analysis (RCA) on the failed bearing to identify primary failure mode (e.g., fatigue, contamination, lubrication starvation, corrosion).', 'Review and optimize CCM lubrication schedules, grease type, and delivery system integrity (pumps, lines, nozzles).', 'Evaluate current bearing design and sealing arrangements for robustness against operating environment (water, scale). Consider upgraded bearing types or improved seals if RCA indicates deficiency.', 'Implement or enhance online condition monitoring (e.g., advanced vibration analysis, acoustic emission) for CCM segment bearings to provide earlier warning of degradation.', 'Review maintenance procedures for CCM segments, including inspection protocols and bearing replacement practices.', 'Train maintenance personnel on advanced bearing fault detection and lubrication system troubleshooting.】

Spare Procurement

【CCM Segment Bearing (specific type and size for the affected segment).', 'Associated seals, spacers, and fasteners for bearing installation.', 'If bearing replacement is part of a larger assembly, a complete spare CCM segment.', 'Ensure adequate stock levels for critical CCM bearings and related components based on failure rates and lead times.】

Executive Summary

A critical CCM Segment Bearing is exhibiting severe symptoms of imminent failure, including high vibration, increasing temperature, and abnormal noise. This poses a high risk of unplanned production downtime, significant equipment damage, and potential safety hazards. Immediate action to replace the affected segment/bearing is imperative to prevent catastrophic failure.

Maintenance Recommendation

It is strongly recommended to initiate an immediate planned shutdown to replace the affected CCM Segment Bearing or the entire segment. Proactive replacement will mitigate the high risk of catastrophic failure, minimize secondary damage, and allow for a controlled repair. A comprehensive root cause analysis must follow the replacement to prevent recurrence.

Conclusion

The presented symptoms unequivocally indicate an advanced stage of bearing failure on the CCM Segment. Failure to address this issue promptly will almost certainly lead to a catastrophic breakdown, resulting in substantial operational and financial impact. Immediate intervention is critical for maintaining equipment reliability, operational safety, and production continuity.